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The American motor parkway

TIMOTHY DAVIS

The development of the American motor parkway was one of the most significant landscape design achievements of the twentieth century. In addition to serving as prototypes for the high-speed motorways that changed the face of the developed world, parkway development provided an international model for the harmonious integration of engineering, landscape architecture, and urban and regional planning (figure 1). At the height of their popularity in the 1920s and 1930s, parkways were championed by the popular and professional press and embraced by a diverse array of interests including engineers, landscape architects, city and regional planners, patriotic societies, scenic beautificationists, public officials, tourism interests and elite architectural critics. Motorists flocked to them as well, cramming the commuter parkways that spread throughout metropolitan suburbs and enjoying extending excursions on the long-distance parkways that traversed America's back-country (figure 2).

The primary reason parkways achieved such widespread acclaim was that they reconciled a number of complex and often competing concerns. Not only were they the most efficient and technologically sophisticated motorways yet developed, but their picturesque aesthetics, rustic allusions and diverse recreational amenities satisfied a wide array of practical and symbolic needs. Parkway began to recede from prominence after 1940 when changing social practices, accelerating technological developments and shifting cultural concerns made it increasingly difficult to accommodate such diverse demands in single multipurpose environments. This article recounts the parkway's rise to prominence, summarises its subsequent marginalisation and speculates on its continued relevance as both an historical artifact and a model for future landscape developments.

From a technical perspective, parkways were clearly the most advanced motorways in the world from the mid-1910s to the mid-1930s. They were safer, faster and more efficient than the alternatives provided by conventional highway builders. On the aesthetic front, parkways afforded a cure for the much-decried problem of roadside blight, while updating longstanding traditions of landscape design to accommodate the needs and speeds of

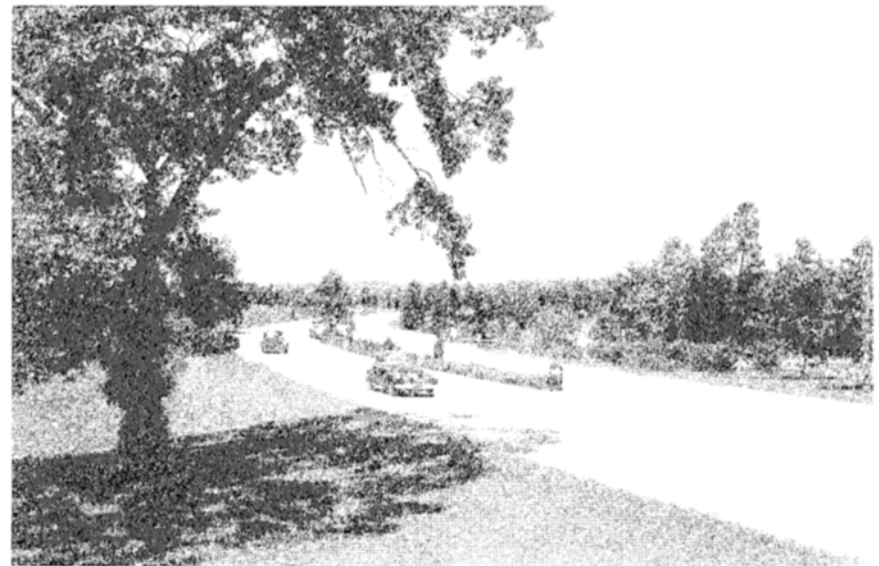


FIGURE 1. George Washington Memorial Parkway, 1946 (United States Bureau of Public Roads Collection, National Archives and Records Administration (NARA)).



FIGURE 2. *Blue Ridge Parkway, c. 1938 (vintage postcard: author).*

contemporary motorists. As public amenities, they provided access to increasingly far-flung parks and beaches, while serving as multipurpose recreational areas in their own right. From a broader planning perspective, they played a vital role in efforts to revitalise rural regions and reshape the form and function of the modern metropolis. Parkway development made good business sense, as well. Parkways proved to be powerful economic engines, gentrifying their immediate surroundings, boosting suburban real estate values across the board and nurturing the increasingly important heritage tourism industry. With their attractive naturalistic scenery and comforting rustic allusions, moreover, parkways were cast as quintessential American landscapes and presented as active agents in the drive to define and promote idealised American values. This was particularly true of federal projects such as the Mount Vernon, Colonial and Blue Ridge Parkways, which were explicitly intended to inculcate a highly selective vision of American history and destiny by celebrating carefully preserved or restored natural and cultural landscapes. While parkways served as repositories of traditional values and tried-and-true picturesque aesthetics, they also captured the imagination of modernist critics and artists, who celebrated their sinuous streamlined curves and hailed them as embodiments of modern conceptions of space, time and motion.⁴

Along with their overt practical values and symbolic appeal, parkways performed an exceedingly important cultural function with broad implications, both in the United States and abroad. By serving simultaneously as icons of modernity and touchstones of traditional values, parkways helped mediate the tension between progress and nostalgia that numerous historians have identified as one of the dominant cultural concerns of the interwar period.² Most Americans embraced the material rewards of modernisation, but few sought a radical break with the past. Popular interest in history — especially colonial and early American history — remained strong as the country looked for a ‘useable past’ to provide continuity, stability and a sense of direction in a rapidly changing world. By combining traditional landscape aesthetics and allusions to America’s pioneer past with state-of-the-art highway engineering, parkways grounded the present in the past, reconciling modern desires for material and social progress with the paradoxical longing to recapture the simple virtues and clarity of purpose commonly ascribed to the pre-modern era. By combining nature, celebratory public history and technological progress in harmonious compositions that could be experienced through the medium of recreational driving, parkways designers united America’s most rapidly growing popular pastime with three of the most prominent themes of American national identity. While the parkway’s practical benefits are more readily documented, a more subtle reason for their widespread appeal was that by being both emphatically modern and resolutely traditional, they helped Americans negotiate the disjunctive experience of modernisation during the tumultuous period between the two world wars.

Parkways were not twentieth-century inventions, of course. Frederick Law Olmsted and his partner Calvert Vaux introduced the term ‘park-way’ to describe the attractive approaches they designed for Brooklyn’s Prospect Park in 1868. The idea of connecting suburban parks with urban centres and elite residential developments by means of attractively landscaped parkways soon became a key element of American city planning. Parkways were seen as means of spreading the benefits of parks throughout urban areas and as powerful economic stimulants that enhanced property values, encouraged high-class residential development and bolstered a city’s reputation as a place for business, residence and tourism. As the connecting fabric of metropolitan park systems, moreover, parkways played an important role in enlarging the scope of American city planning from isolated projects to comprehensive improvements.³

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Stylistically, the first parkways resembled traditional boulevards. Brooklyn's Eastern Parkway consisted of a central pleasure drive bordered by formally arrayed tree-lined park strips, which were flanked by additional roadways for commercial vehicles and access to abutting properties. The series of drives and bordering parkland that Olmsted planned for Boston's Muddy River in the 1880s (now known collectively as 'The Emerald Necklace') redefined the parkway concept from a formal avenue to a picturesque lane winding through an elongated park embellished in the naturalistic style favoured by Anglo-American landscape designers. The Boston project also represented a middle stage in the evolution from traditional boulevard to modern parkway in circulation terms. The riverside location limited access on one side of the main roadway, but turning and entering traffic from the adjacent streets remained a source of danger and disruption. Despite improvements, the advent of the automobile rendered traditional parkways obsolete on both practical and aesthetic grounds. Not only were the tight turns and frequent intersections unsuited for automobile traffic, but the intricate picturesque compositions favoured by nineteenth-century landscape architects became increasingly difficult to apprehend as speeds rose from a leisurely horse-drawn pace to a heart-pounding 30 to 40 miles per hour.

The primary impetus for the parkway's rising popularity had less to do with refined aesthetic arguments than with pragmatic concerns for safety and efficiency, augmented by a broadly shared contempt for the excesses of commercial roadside development. Conventional highway departments made significant progress in basic engineering issues, alleviating the 'dust menace' and 'getting the farmer out of the mud' through the adoption of modern paving methods. By the 1920s, however, it was apparent that simply paving over old roadways was not an adequate solution to America's highway crisis. The chief problem with conventional roadways, most critics agreed, was the combination of poorly designed intersections, hazardous and unsightly road alignments, and rampant and unconstrained commercial development. Exacerbating the dangers posed by at-grade intersections and unregulated side streets was the longstanding legal precedent that guaranteed landowners access to roadways passing by or through their property. With the growing popularity of the automobile, the guaranteed frontage right rules were turning America's roadways into congested, unsightly and dangerous linear slums (figure 3). Merchants lined the highways with brightly coloured billboards and constructed a chaotic array of roadside establishments to capitalise on the

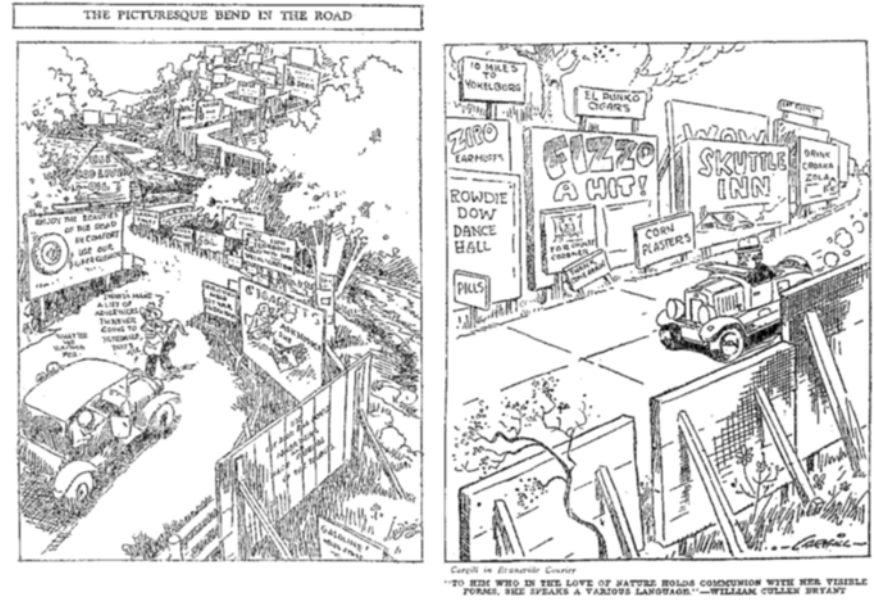


FIGURE 3. Typical anti-roadside blight cartoons (reprinted in *American Civic Annual, 1929* and *American City, 41* (1929)).

burgeoning market of motoring Americans. The unregulated entrances to these establishments disrupted the flow of through-traffic, while the eye-catching signs and garish architecture distracted motorists and disappointed sightseers in search of attractive rural scenery. The general consensus was that outmoded and unconstrained highway development was ruining the traditional appearance of the countryside, stifling urban and suburban growth, and preventing Americans from enjoying the full potential of the automobile age.

What was needed, in the eyes of contemporary planners and popular critics, was a new type of roadway geared to the social and technological requirements of modern, automobile-dominated life. Professional journals were filled with prescriptions for highway improvements, scenic improvement groups excoriated the billboard and roadside merchandising industries, and middle-brow magazines published polemics about the dangers and disappointments of the American roadside. *Fortune* magazine quoted academic experts on the dangers of modern highway travel; *Life* magazine complained that the

American roadside had become 'the supreme honky-tonk of all time'; and the *Saturday Evening Post* called for 'a new and happier era of highway construction' in which engineers and landscape architects joined forces to 'make motor travel as stimulating to the eye as it is to the speedometer'. Benton Mackaye and Lewis Mumford used the pages of *Harpers* and the *Atlantic Monthly* to call for a new approach to highway development based on 'motor age principles'.⁴

By the mid-1920s, it was becoming increasingly apparent that the most significant progress in developing this new type of roadway was being made in the rapidly expanding network of parkways spreading from New York City into Westchester County and Long Island. By combining traditional parkway design techniques with new technologies and adapting them to higher speeds and longer distances, designers such as Jay Downer, Gilmore Clarke and Hermann Merkel invented a new type of roadway that accommodated modern automobile traffic without sacrificing the parkway's traditional scenic,

recreational and environmental functions. The new parkways along the Bronx, Sawmill and Hutchinson Rivers were flanked by wide tree-lined borders that greatly reduced the number of accident-prone exits and entrances while screening out unsightly billboards and other roadside developments (figure 4). The safety and efficiency of these 'limited-access' parkways was enhanced by grade separations that eliminated cross traffic at busy intersections (figure 5). Parkway alignments followed graceful s-curves and rose and fell gently in harmony with the surrounding terrain. This made driving more enjoyable while minimising unsightly excavations, reducing hazardous curves and eliminating dangerous blind-spots. Careful attention to the three-dimensional quality of road design resulted in improved sight-lines and eliminated the dangerous and unpleasant roller-coaster effect of climbing up and down hills in rapid succession. Following existing contours also reduced the need for unsightly cuts and fills that scarred the sides of conventional roadways. While landscaped median strips were employed in select locations to preserve particularly attractive terrain, the expense of constructing separate roadways kept this clearly desirable safety feature from being employed on a

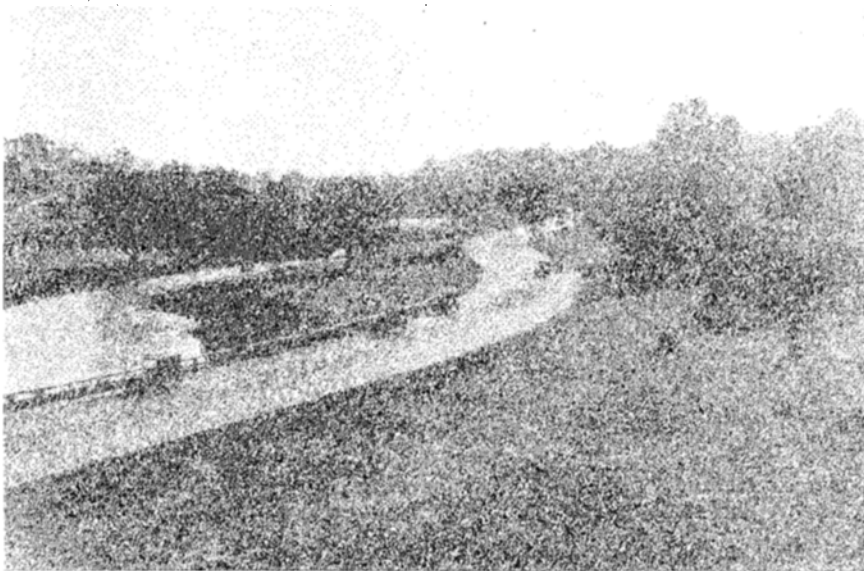


FIGURE 4. Classic Bronx River Parkway view illustrating broad landscaped corridor, naturalistic grading and serpentine alignment, 1922 (Report of the Bronx Parkway Commission, 1922).

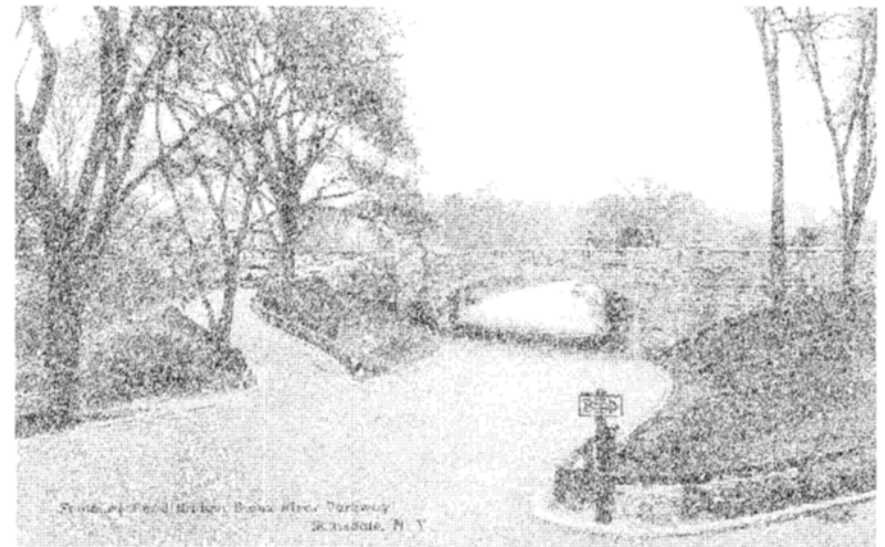


FIGURE 5. Classic masonry-faced grade separation structure eliminating dangerous intersection, Bronx River Parkway, c. 1925 (vintage postcard: author).

wide scale during the first decade of motor parkway development. The modern concrete bridges and grade separations that were essential to the parkway's safety and efficiency were given rustic surface treatments and integrated with the surrounding terrain with the help of picturesque plantings. Together with the rustic guard rails and light posts and the carefully composed naturalistic landscaping, these attributes tempered the striking modernity of the parkway's sophisticated circulation features with reassuring allusions to vernacular traditions and classic landscape aesthetics.⁵

The Bronx River Parkway was the first of these new parkways to be completed; opening with considerable fanfare in November 1925 (figure 6). The Bronx River Parkway proved to be tremendously successful, both as a scenic pleasure drive and as a commuter thoroughfare. With the new parkway providing convenient access to New York City, formerly remote areas of Westchester County experienced a tremendous real estate boom. This parkway-driven prosperity stimulated realtors and civic boosters to join forces with park promoters to advocate the construction of attractively landscaped, limited-access parkways throughout the region. Soon Westchester County and southern Long Island were in the midst of a parkway-building frenzy.

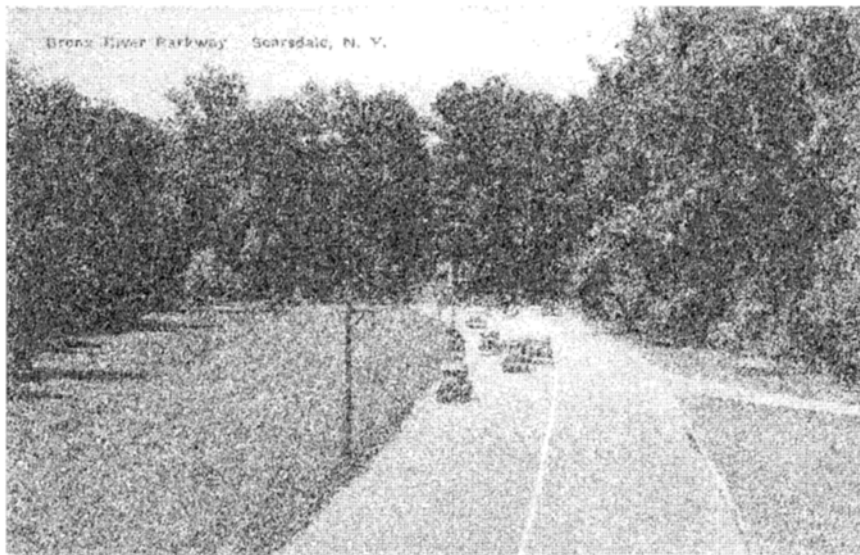
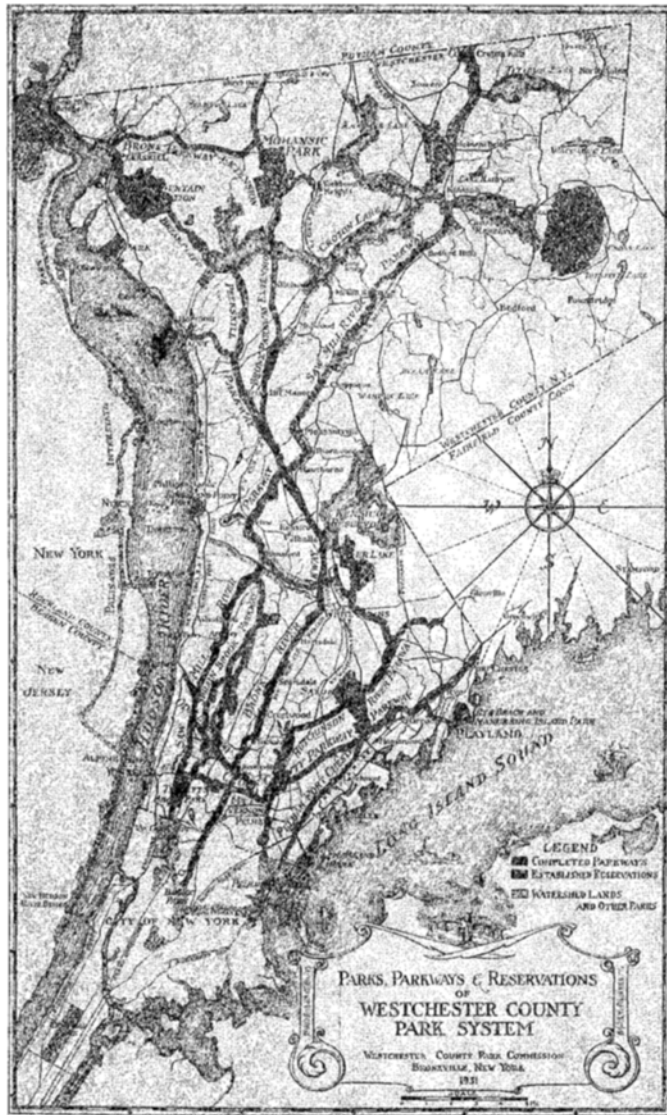


FIGURE 6. *Bronx River Parkway, c. 1925 (vintage postcard: author).*

Westchester County alone constructed another 70 miles of parkway over the next twenty years (figure 7). These additions included the Saw Mill River Parkway, the Hutchinson River Parkway and the Cross County Parkway. New York park czar Robert Moses, meanwhile, pressed for construction of a series of parkways heading from New York City to the beaches and parks he was developing on Long Island. The Southern and Northern State Parkways and their various feeder routes employed many of the design features that characterised their Westchester County contemporaries (figures 8 and 9).

While these parkways were widely heralded as the safest and most efficient motor roads in America, they were developed by park commissions and intended to function as mixed-used recreational environments. They were conceived and constructed as integral components of ambitious local and regional park systems. The gleaming modern motorways were often flanked by bridle paths and walking trails (figure 10). At select locations, the parkway corridors broadened to include community parks, golf courses, lakes, athletic fields and other recreational facilities (figure 11). Many led to state parks and beaches that were heavily used by urban and suburban residents. Another key element of traditional park and parkway design — the prohibition of buses, trucks and commercial vehicles — ensured a more relaxed and attractive driving environment and allowed designers to work with narrower roadways and more sinuous curvature.⁶ Most of these parkways were designed by teams of landscape architects, architects and engineers working in close collaboration; with landscape architects generally playing the leading role in matters of road layout and landscape design (figure 12).

Despite the success of the Bronx River Parkway and its fellow Westchester County and Long Island parkways, the highway engineering community did not rush to embrace parkway design principles. Downer, an engineer, and Clarke, a landscape architect, made repeated appeals to the highway-building fraternity, presenting the Westchester County parkways as examples of the fruitful collaboration of landscape architects and engineers. In speeches and professional journal articles, they insisted that the parkways' aesthetic attributes presented significant practical advantages as well. Sinuous alignments were safer than the long straight segments linked by short-radius curves favored by highway engineers. Gently sloped and rounded banks resisted erosion better than the steep raw cuts left by traditional construction methods. Saving topsoil and replanting roadsides with native vegetation reduced costs and produced a more stable, maintenance-free environment. Senior officials of the United



Courtesy of Westchester County Park Commission

FIGURE 7. Map of Westchester County Park Commission parks and parkways, 1931 (courtesy of Westchester County Records and Archives Center).

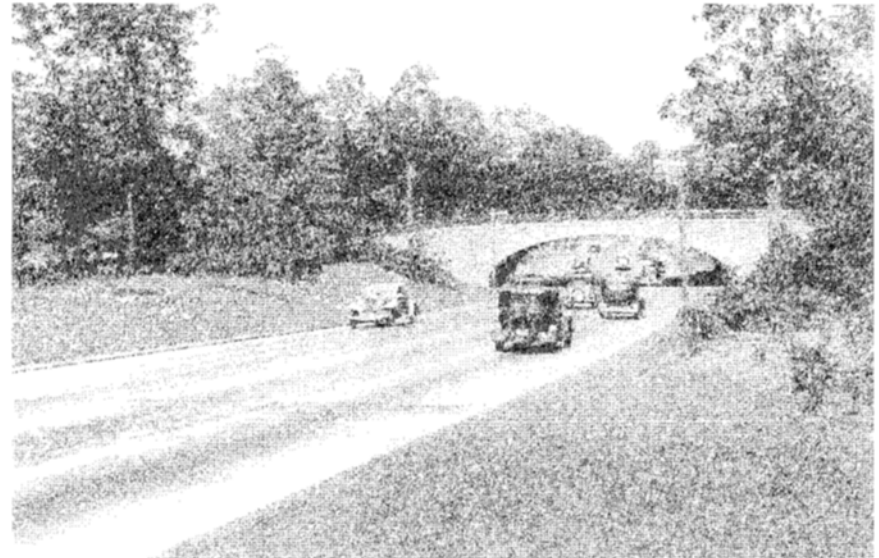


FIGURE 8. Typical Long Island State Parkway scene, c. 1932 (Long Island State Parkway Commission).

States Bureau of Public Roads (BPR) quickly recognised that the new parkways were not just more attractive than ordinary roadways, but safer and more efficient as well. The problem, however, was to convince skeptical state highway engineers, budget-strapped planners, tight-fisted politicians and the broader American public that parkway-style design principles could be translated to large-scale highway development. Most engineers and highway officials viewed landscape architects as extravagant artistes intent on driving up the costs of roadway development with gratuitous ornamentation. Parkway might be appropriate for limited recreational developments in well-heeled suburbs, they believed, but accommodating the landscape architects' desire for serpentine curves and broad landscaped rights-of-way was as alien to the engineers' culture of mathematical precision and economic calculation as acceding to the whims of female-dominated scenic beautification groups, with whom landscape architects were often linked in the professional mindset.⁷ Both constituencies were marginalised as dilettantes and 'prettifiers' who had no business meddling with the serious masculine business of large-scale highway development.

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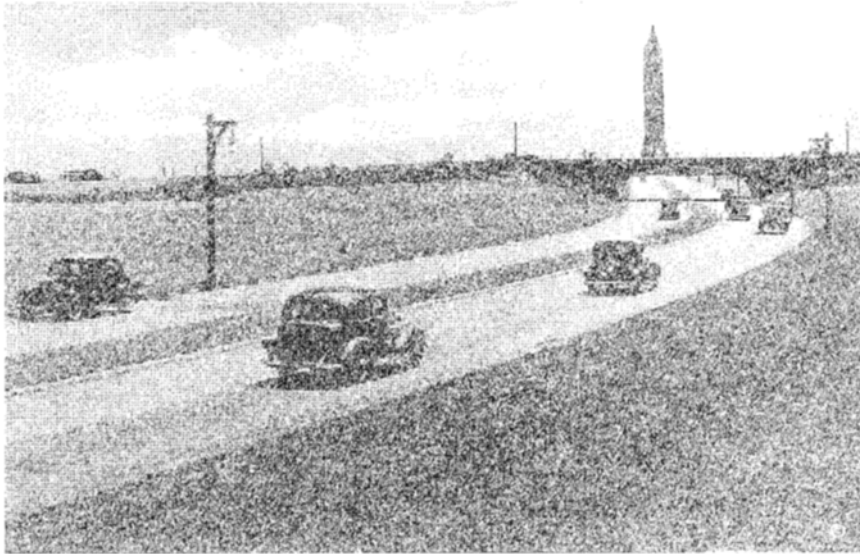
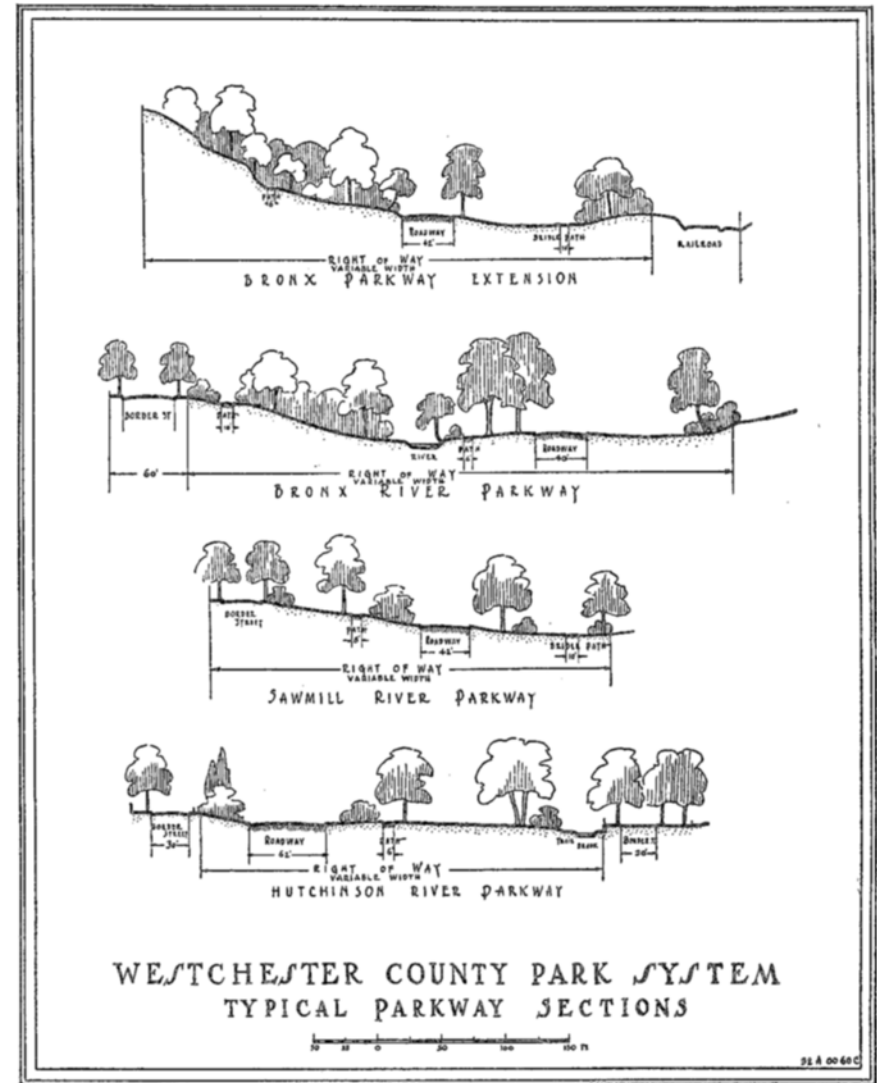


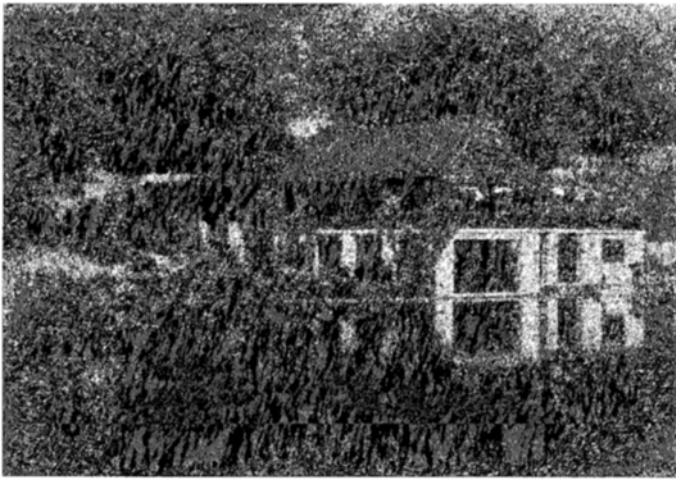
FIGURE 9. *Wantagh Parkway approaching Jones Beach, Long Island, c. 1938 (vintage postcard: author).*

The long-awaited construction of a memorial roadway from Washington to Mount Vernon provided BPR officials with an ideal forum for convincing skeptical rank-and-file engineers that parkway-style development techniques could be applied to mainstream arterial highway construction in an efficient and economical manner. When the BPR was put in charge of the project in 1928, the agency went to great lengths to present Mount Vernon Memorial Highway as a 'model modern motorway,' casting it as a national — and even international — paradigm of modern motorway design. Even the name 'Mount Vernon Memorial Highway' was calculated to further the goal of convincing mainstream engineers to adopt parkway-style design principles. As a tourist-oriented roadway restricted to private vehicles and surrounded by a wide band of multipurpose parkland, the project clearly fit the conventional definition of a parkway. Calling it a memorial highway and promoting it with a barrage of publicity in mainstream highway engineering circles maximised the project's potential to convince conservative highway engineers of the practical value of parkway-style development.



Courtesy of Westchester County Park Commission

FIGURE 10. *Westchester County Park Commission parkway sections, c. 1934 (courtesy of Westchester County Records and Archives Center).*



Restaurant at V. Everit Macy Park, Saw Mill River Parkway, Westchester County Park System
Gilmore D. Clarke, Landscape Architect



Underpass Bridge for Bridle and Foot Path, Hutchinson River Parkway, New Rochelle,
Westchester County Park System
Gilmore D. Clarke, Landscape Architect. Courtesy Office of Public Buildings and Public Grounds
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FIGURE II. Recreational developments along Westchester County parkways (Parks and Recreation, March 1933; courtesy of United States Department of the Interior Library).

The BPR prepared articles on the parkway's design, conducted tours for visiting highway officials, and produced impressive booklets and detailed technical pamphlets outlining the project's design features. Grading plans, cross-sections and circulation diagrams emphasised the practical value of free-flowing traffic provisions and underscored the harmonious integration of landscape architecture and highway engineering (figure 13). Compelling before, during and after photographs delineated the construction process and documented the transition from tangled woodland or cluttered roadside to attractive and efficient parkway landscape. The most elaborate of these publications was the 1934 United States Department of Agriculture (USDA) bulletin *Roadside Improvement*. The Nazi highway engineer Fritz Todt found this document so impressive that he reprinted it in German to help guide development of the *Reichsautobahnen* (figure 14).⁸

The memorial highway garnered worldwide attention when the BPR hosted an international road conference in Washington in October 1930, using the project to showcase American achievement in motorway development. The BPR conducted tours of the construction site and published a detailed brochure loaded with attractive graphics. Excerpts from this publication and related reports appeared in international journals such as the Italian highway construction magazine *Strade*.⁹ The German road-building journal *Die Strasse* published numerous accounts of American parkway development during this period. Nazi highway engineers even traveled to Westchester County to inspect the American parkways. Gilmore Clarke hosted the visitors and later maintained that the Germans 'took our ideas' as the basis for the *Reichsautobahn*. Clarke contended that the Nazi designers paid little attention to the more subtle details and broad-based goals of parkway development, however, producing a more efficient, but aesthetically inferior, product.¹⁰

As part of the BPR's efforts to celebrate its achievements and promote parkway design principles, the agency produced a thirty-minute film singing the parkway's praises and portraying the development process in meticulous detail (figure 15).¹¹ Toward the end of the film, the parkway's broader appeal was underscored in a dynamic sequence depicting late-model automobiles speeding along the shores of the Potomac (figure 16). The technical aspects of the film were clearly geared toward a professional audience, but these scenes of speeding automobiles sweeping along intersection-free concrete roadways, with long sinuous curves winding through attractive woodlands, undoubtedly appealed to the legions of motorists who embraced the harmonious integration

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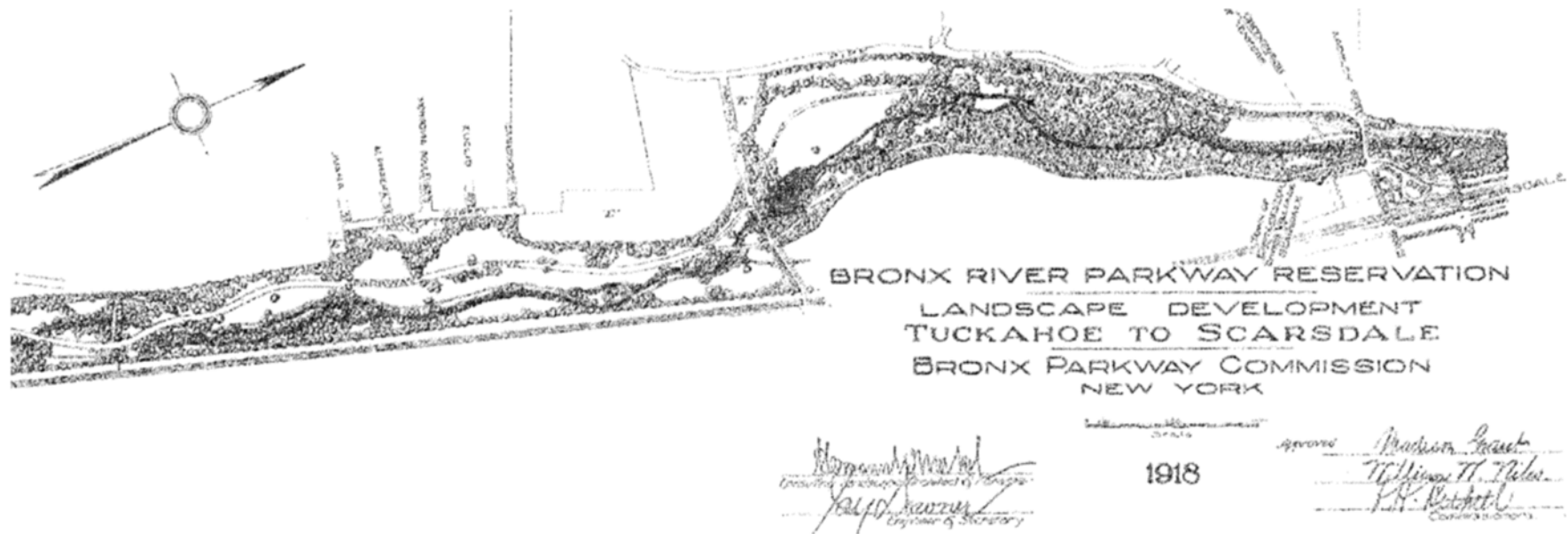


FIGURE 12. Bronx River Parkway plan detail (courtesy of Westchester County Records and Archives Center; original image digitally enhanced for legibility).

of modern roads, modern cars and naturalistic scenery as the ultimate manifestation of America's ability to unite reverence for nature and traditional values with social and material progress.

While the BPR promoted the memorial highway as 'America's Most Modern Motorway', the agency went to great lengths to call attention to the project's historical associations and ideological implications. Not only did the memorial highway link sites of surpassing national significance, but the route itself was designed to encompass landmarks associated with George Washington and other founding fathers. Iconographically significant design details such as broken-pediment Colonial Revival-style signboards underscored the parkway's symbolic associations, as did the Colonial Revival concession stand at the Mount Vernon terminus (figure 17). The drive from Washington to Mount Vernon was presented as a patriotic undertaking with wide-ranging cultural implications. In addition to appreciating the technical superiority of American highway engineering and enjoying the parkway's scenic beauty and opportunities for healthful outdoor recreation, motorists were expected, in the words of one supporter, to experience 'a thrill of

Americanism' while zipping along the shores of the Potomac at the exhilarating rate of 35 mph.¹²

The engineering achievement of constructing the memorial highway was presented as a patriotic achievement in its own right. Wilbur Simonson and others repeatedly described the project as fitting memorial to George Washington, asserting that the founding father's surveying background would have predisposed him to appreciate the parkway's technical excellence. Taking advantage of the project to emphasise the national significance of road-building, the American Association of Highway Officials timed its annual meeting to coincide with the memorial highway's opening celebration. *American Highways* magazine commemorated this event with a striking cover photograph of steadfast engineers assembled under Mount Vernon's iconic portico. The back cover further underscored the patriotic associations of highway engineering with a woodcut of George Washington in surveyor's garb, ostensibly enacting the role of America's first highway builder. The BPR's film heralded the parkway's completion, not just as a memorial to George Washington, but as a demonstration of 'the progress made in highway

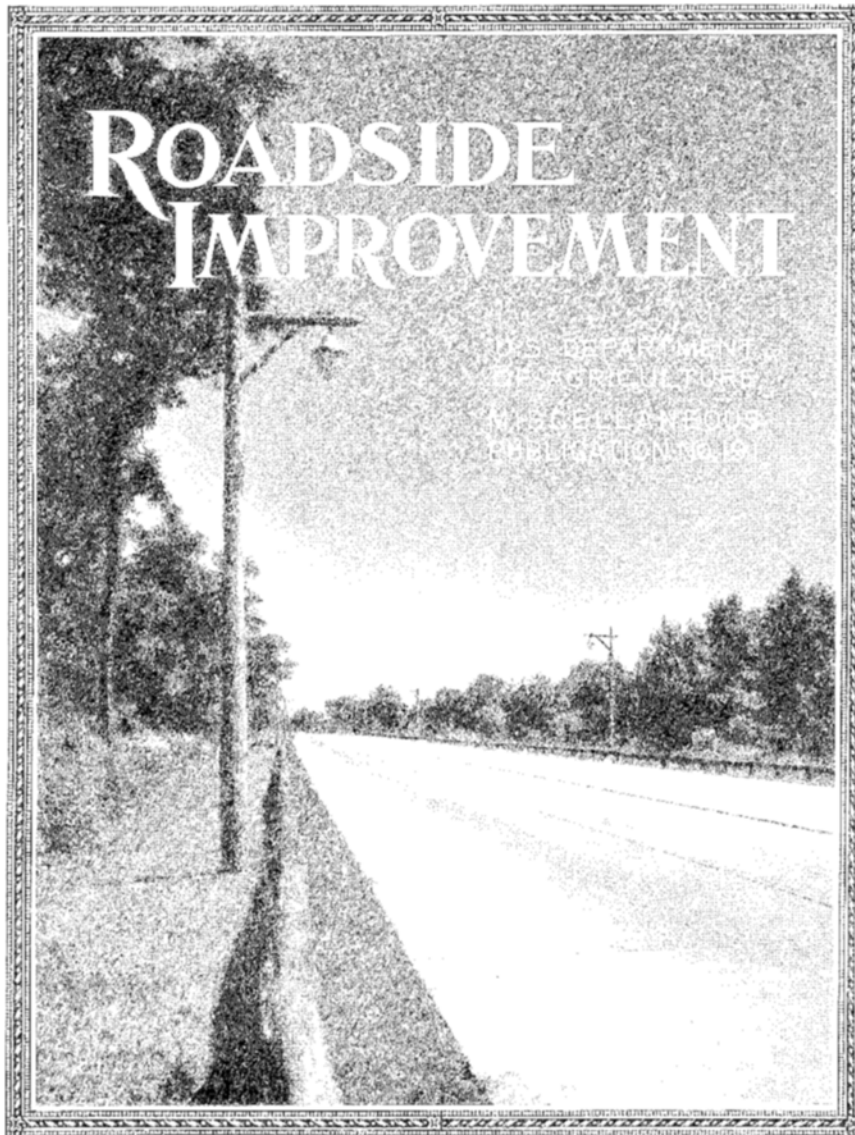


FIGURE 13. One of the many technical publications based on Mount Vernon Memorial Highway (United States Department of Agriculture (USDA) Roadside Improvement (Washington, DD, 1934; author's collection)



FIGURE 14. Cover of contemporary German reprint of Bureau of Public Road's Roadside Improvement bulletin (Deutsches Museum).

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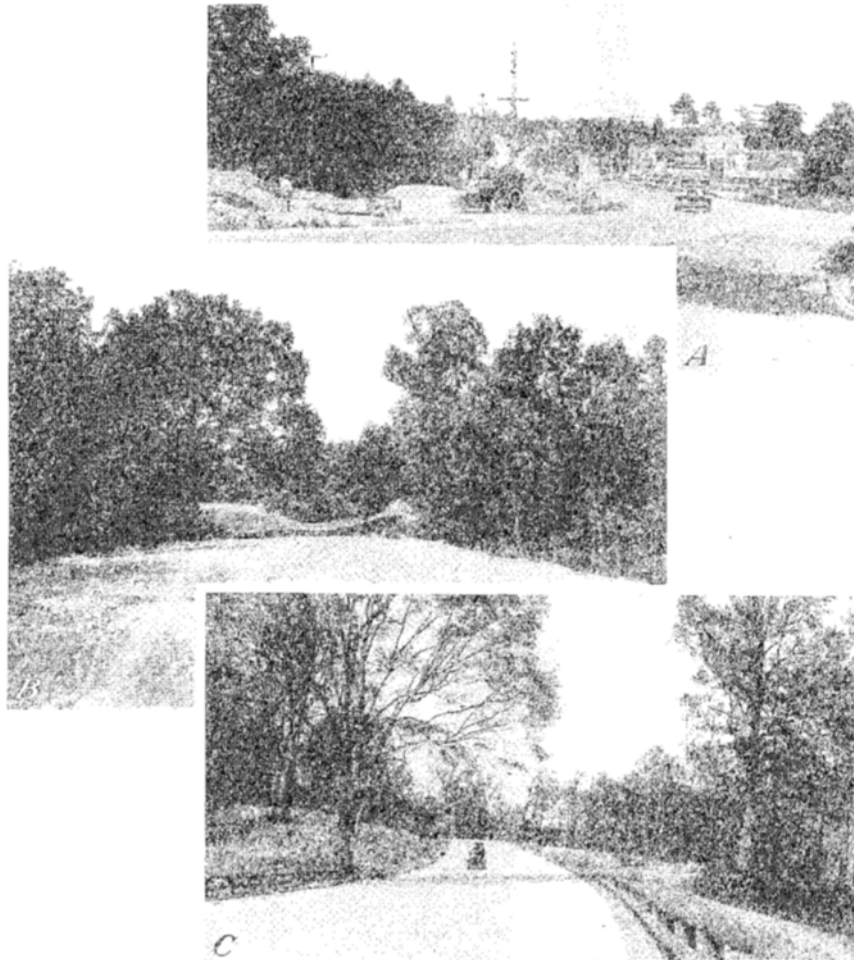


FIGURE 15. Stages in the creation of Mount Vernon Memorial Highway (USDA) Roadside Improvement (Washington, DC, 1934; author's collection).
A, THE APPROACH TO WASHINGTON OVER THE OLD ROAD FROM THE SOUTH; B, VIRGIN TERRITORY WITHIN A SHORT DISTANCE OF THE FIRST PICTURE AND ON THE NEW LOCATION OF THE MOUNT VERNON MEMORIAL HIGHWAY; C, THE COMPLETED HIGHWAY AT SAME LOCATION NOW USED BY PASSENGER VEHICLES APPROACHING WASHINGTON.

FIGURE 15. Stages in the creation of Mount Vernon Memorial Highway (USDA) Roadside Improvement (Washington, DC, 1934; author's collection).

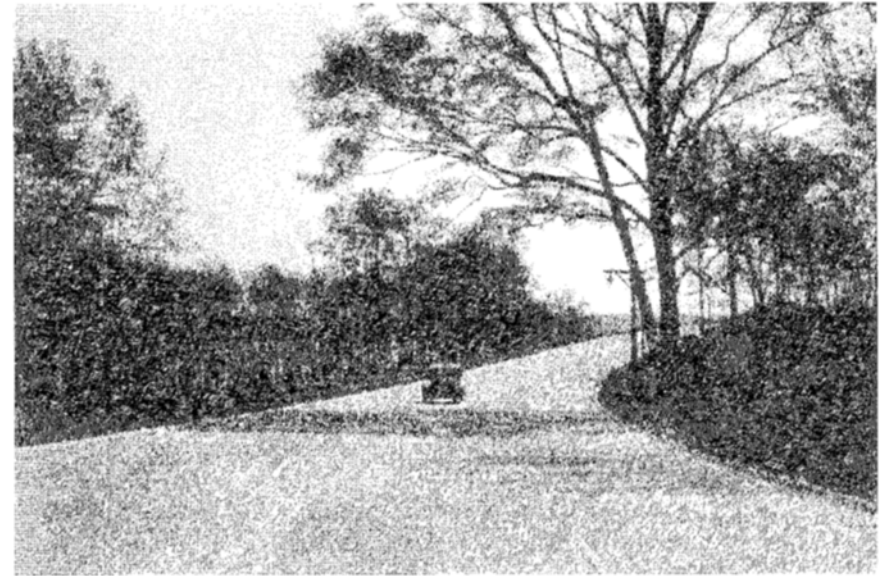


FIGURE 16. Newly completed Mount Vernon Memorial Highway, 1932 (NARA).

engineering' since the birth of the republic. Elaborate models of the parkway's Mount Vernon terminus and cloverleaf interchange were displayed in the Capitol Rotunda in Washington, DC, surrounded by paintings of momentous events in American history.¹³

The popular and professional press heralded the project as an exemplary union of modern motorway development, landscape design and patriotic sentiment (figure 18). Articles on the memorial highway's design, construction and cultural significance appeared in newspapers, popular magazines and virtually every major planning, engineering and park development publication. Mainstream engineering periodicals echoed the BPR's pronouncements about the practical value of parkway-style design, providing detailed technical information and invoking the BPR's characterisation of the project as 'America's Most Modern Motorway'.¹⁴ Landscape architects were equally enthusiastic, applauding the memorial highway for demonstrating the value of cooperation between engineers and landscape architects.¹⁵ General publications focused on the parkway's recreational potential, scenic beauty, historical associations, and commemorative significance. *Literary Digest* praised the



FIGURE 17. Colonial Revival signs and concession stand, Mount Vernon Memorial Highway, 1932 (NARA).

AMERICAN MOTORIST



FIGURE 18. Cover of American Automobile Association magazine celebrating completion of Mount Vernon Memorial Highway, 1932 (Library of Congress).

project as 'a yardstick among highways'; proclaiming that the new roadway would be 'as pleasing to the eye as it is to the rumble seat'.¹⁶ Explicitly invoking the themes of nature, patriotism and technological prowess, *American Motorist* applauded the BPR for having 'perpetuated Washington's memory by a boulevard of which he would be proud from a practical, patriotic, and picturesque point of view'. Reiterating these sentiments in a formulation that underscored the cultural resonance of road-building, the magazine enthused: 'At last there is a highway built for beauty with history for its roadbed and the American ideal for its goal.'¹⁷

While the National Park Service (NPS) was not directly involved in the development of Mount Vernon Memorial Highway, the agency embraced the notion that parkways could serve as linear lessons in American history and values. Colonial Parkway, begun in 1930 and largely completed in the 1930s, even more explicitly catered to the period's veneration of the early American past. Modern concrete bridges were clad in red-brick veneer, the sinuous modern pavement was subtly textured to mimic the rough surfaces of Colonial-era roads, and the route connected three of Virginia's most venerated historic sites: Jamestown, Yorktown and Williamsburg (figure 19). The



FIGURE 19. Colonial Parkway, c. 1935 (vintage postcard: author).

parkway served as the primary circulation for Colonial National Historical Park — one of the NPS's first large-scale ventures into historic site management and heritage tourism. As with most public history enterprises of this era, the NPS's initial efforts supported traditional sagas of 'great men and events' by preserving presidential homes, battlefields, and locations associated with momentous developments such as the first permanent Anglo-Saxon outpost in the New World and the surrender of British forces that sealed the success of the American Revolution.

The Blue Ridge and Natchez Trace parkways were meant to tell a more populist story; celebrating America's agrarian roots and preserving relics of the nation's hardy pioneer past. Along with providing recreational opportunities and presenting spectacular views that reaffirmed the longstanding belief that America was 'Nature's Nation' (figure 20), these long-distance rural parkways were explicitly intended, in the words of NPS landscape architect Stanley Abbott, to 'provide the look of homespun in an east that is mostly silk and rayon'.¹⁸ Stretching for hundreds of miles through backcountry Virginia, North Carolina, Mississippi, Alabama and Tennessee, the Blue Ridge and Natchez Trace were linear odes to America's agrarian past. With miles and miles of sleepy pastures, split rail fences and quaint log cabins, they afforded motorists with soothing respites from the pressures of modern urban life (figure 21).

These pastoral idylls were as contrived as any English landscape park or romantic painting. While BPR engineers attended to the technical aspects of parkway-building, NPS landscape architects composed carefully calculated scenes of rural bliss. Abbott and his colleagues referred to this enterprise as improving the 'roadside picture' and evidenced little concern for the literalistic notions of historical accuracy that would come to dominate later public landscape debates. Buildings that failed to conform to pioneer stereotypes were torn down, charismatic structures were moved to more accessible locations and new construction emulated regional vernaculars. Traditional agricultural practices were promoted through leases and scenic easements. While parkways were ostensibly intended to celebrate traditional folk values, the actual folk were not accorded the most respectful treatment. Parkway development displaced significant numbers of rural residents, though some elderly individuals were granted lifetime tenures to enliven the NPS's drive-through dramatisation of Arcadian America. NPS officials appeared untroubled by the social cost of parkway development to the local populations, many of whom

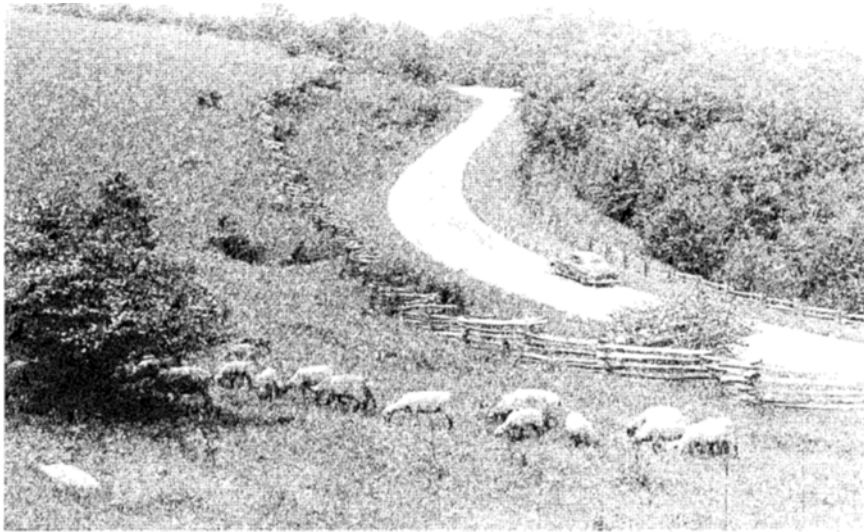


FIGURE 21. *Blue Ridge Parkway, c. 1954 (National Park Service).*

had lived on the land for generations. The displacement of a few 'backward mountaineers' was seen as a small price to pay for the broader good of enabling millions of Americans to experience a sense of cultural regeneration by driving along a roadway designed to memorialise the nation's mythic roots.¹⁹

This combination of recreational motoring and aggressively nationalistic heritage tourism was not unique to American parkway development, by any means. Nazi autobahn developers and political leaders similarly believed that encouraging interregional travel and showcasing highly selective natural and cultural landscapes would promote a shared national culture based on idealised folk values and the redemptive power of nature and rural life. Nor were the Germans and Americans alone in this enterprise, though few other countries could devote similar resources to the development of state-sponsored tourist infrastructures. As numerous historians of the late nineteenth and early

twentieth centuries have demonstrated, cultural leaders and ordinary people throughout the developed world were engaged in a process of defining, redefining, inventing and manipulating various aspects of their real or imagined heritage, both to counter the social dislocations and uncertainties wrought by modernity and to strengthen the power and identity of the nation-state. These activities ranged from renewed interest in folk tales, ballads and 'traditional' crafts, to the staging of pageants and festivals, the development of 'living' museums, the celebration of vernacular architecture and the alignment of local, regional and national political agendas with putatively indigenous cultural ideals. This 'romantic nationalism', as it has been called, was particularly prevalent in the United States, Britain and northern Europe, where it strongly influenced popular culture and 'high art' and appealed both to outright conservatives threatened by the changing social and racial dynamics of modern life and to putative progressives who sought to combine traditional values with advances in various social and technological spheres.²⁰

Parkways embodied both strains of romantic nationalism. The celebration of Anglo-American history and folk values soothed contemporary anxieties about the impacts of immigration and urbanisation on American life, while the gleaming concrete roadways winding through natural scenery and traditional agrarian landscapes epitomised the reconciliation of Modern and Anti-Modern impulses. By reinterpreting the winding country road and the iconographic American landscape through the quintessentially modern medium of the streamlined motorway, parkway designers provided a populist counterpoint to the work of avant-garde artists such as Aaron Copeland and Martha Graham, who similarly sought to express traditional idioms and values through modern modes of expression.

The rapidly growing network of suburban commuter parkways was even more dramatically modern, and was celebrated as such by a wide variety of contemporary commentators (figure 22). As metropolitan New York continued its parkway-building binge, the British art magazine *The Studio* proclaimed that Moses' Long Island parkways offered a preview of 'The New York of Tomorrow' that promised 'a new way of living and working' based on 'a fusing of the arts and technical skills and close collaboration with nature'. Thanks to Moses, the magazine declared, 'the highway has become munificent and a thing of beauty.'²¹ *Fortune* magazine asserted that parkways had joined New York's fabled skyscrapers as icons of progress that visitors to the world's most modern metropolis 'simply must see'.²²

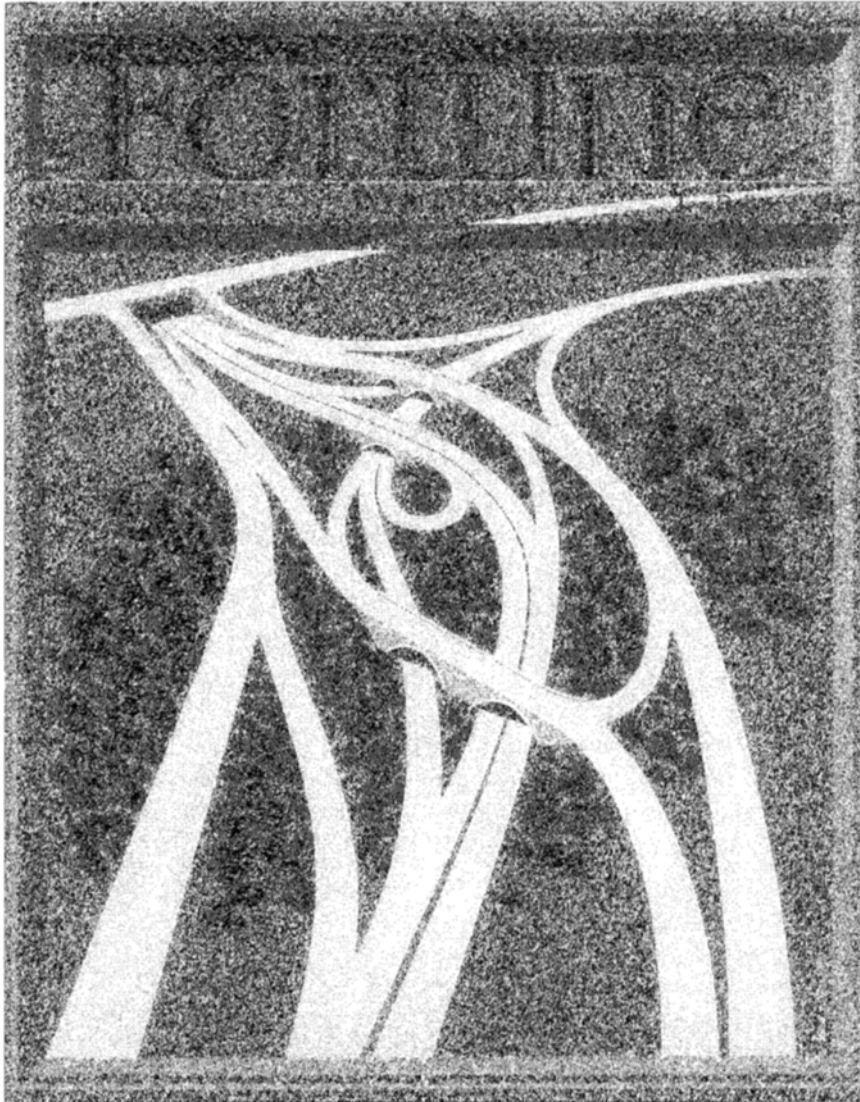


FIGURE 22. Fortune magazine cover celebrating Long Island Parkway interchange, 1938 (author's collection).



FIGURE 23. Postcard with notation extolling New York City parkways (author's collection).

While it is difficult to recapture popular perspectives on parkway development, sentiments expressed on postcards provide some sense of the vivid impressions these new motorways made on their users. A California visitor to Westchester County scrawled: 'Everything is Parkways back in New York' on the front of postcard of Bronx River Parkway that portrayed the gleaming roadway nestled between rugged timber guardrails and soothing greenery (figure 23). On the back he observed: 'I though L.A. and vicinity had a lot of paved roads & highways, but I never saw anything like N.Y. & vicinity.' Underscoring the concomitant importance of the parkway's naturalistic landscaping and rustic features, the anonymous writer praised the scene's 'quaint' appearance as well.²³

Architectural critic Sigfried Giedion extolled the practical and aesthetic value of American parkways. Giedion was particularly impressed by the way in which parkways exemplified the 'space-time' continuum he identified with Modernist perception. Praising 'the great sweep of the highway, the beauty of its alignment, the graceful sequence of its curves', Giedion proclaimed: 'The space-time feeling of our period can seldom be felt so keenly as when driving,

the wheel under one's hand, up and down hills, beneath overpasses, up ramps, and over giant bridges.²⁴ For less esoterically inclined commentators, the most common rhetorical device for emphasising the parkways' modernity was to align them with streamlined automobiles and other emblems of modern industrial design. Mount Vernon Memorial Highway landscape architect Wilbur Simonson enthused: '[T]his broad paved highway will simulate in its flowing lines, the spiral curves, the horizontal and vertical transitions, and the banked turns of a fast transport aircraft in flight.'²⁵ Not only was 'streamlining' one of the dominant popular symbols of modernity, but the association had legitimate technical underpinnings as well. Traffic experts had developed elaborate theories ascribing automobile accidents to the 'friction' produced by poorly designed roadways, hazards that parkway designers literally 'streamlined' away.²⁶ Simonson insisted that "'streamlining" the flow of high speed modern motor traffic' with teardrop safety islands, spiral curves, beveled curbs, gently rounded side-slopes and outwardly flared guard rails exemplified the designer's determination to ensure that 'the practical needs of traffic circulation and aesthetic attractiveness are harmonized'.²⁷

Even as the classic 1930s motor parkways were being heralded for their masterful integration of naturalistic landscape aesthetics, technological progress and traditional American values, these competing concerns were starting to pull the disparate elements of the parkway movement apart. Changing cultural and technological factors made it increasingly difficult, if not impossible, for a single environment to accommodate the conflicting demands for higher speeds and larger traffic volumes, extensive reaches of undeveloped scenery and increasingly elaborate public history presentations. Not only did efficiency-minded highway engineers realise that they could appropriate the parkway's path-breaking circulation features without expending money on scenic improvement and associated recreational developments, but landscape architects and environmentalists were coming to the conclusion that motor road development was incompatible with evolving park and preservation ideals. The multipurpose parkways epitomised by Mount Vernon Memorial Highway gradually gave way to more specialised environments: urban expressways, high-speed toll roads, remote scenic parkways and an ever-broadening array of recreational areas, wilderness preserves and historical parks.

By the mid-1930s, the parkway movement was clearly diverging in two directions: the relatively low-speed scenic and recreational parkways promoted by the National Park Service, and the commuter parkways proliferating

around major American cities. Connecticut's Merritt Parkway was the mostly widely celebrated of this new generation of commuter parkways. It was heralded as both a marvel of modern motorway development and an engaging example of landscape design. Despite its tree-lined borders and prolific plantings, however, the Merritt was decried by parkway purists on the grounds that the relentless straight-ways and lack of associated recreational features made it more of a glorified traffic artery than a sensitively designed multipurpose recreational development (figure 24). Los Angeles's first true modern motorway epitomised the parkway's changing fortunes. The Arroyo Seco Parkway was conceived as a classic recreational amenity, but by the time it was completed in the early 1940s it had become a stripped-down traffic artery. Its new name — the Pasadena Freeway — heralded the changing form and function of urban motorways. Freeways (the term was coined in 1930 by E. M. Bennett to denote freedom of movement, not the absence of tolls) would reshape the American landscape in the postwar era with a distinctly different balance between efficiency and landscape aesthetics.²⁸ Higher speeds and larger traffic volumes demanded wider, straighter roads that were difficult to reconcile with traditional landscape values. As the Nazi regime began to complete and showcase significant stretches of the *Reichsautobahnen* in the

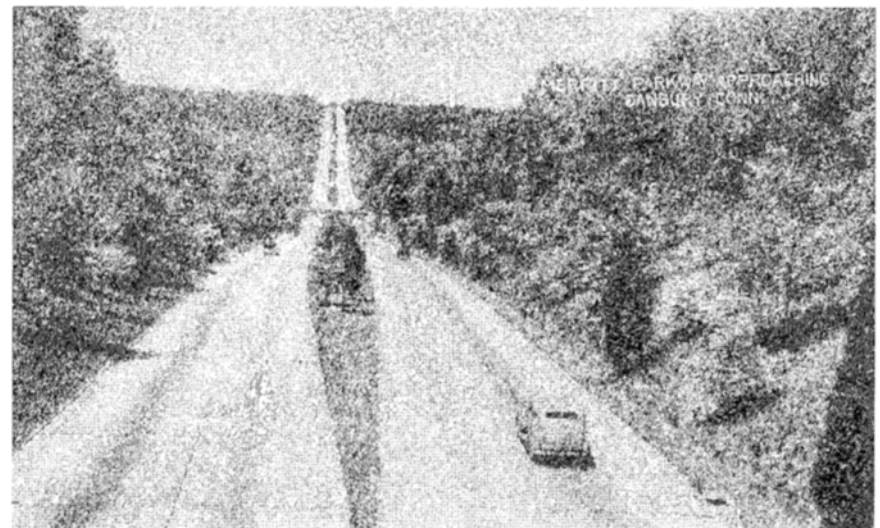


FIGURE 24. Merritt Parkway, c. 1938 (vintage postcard: author).

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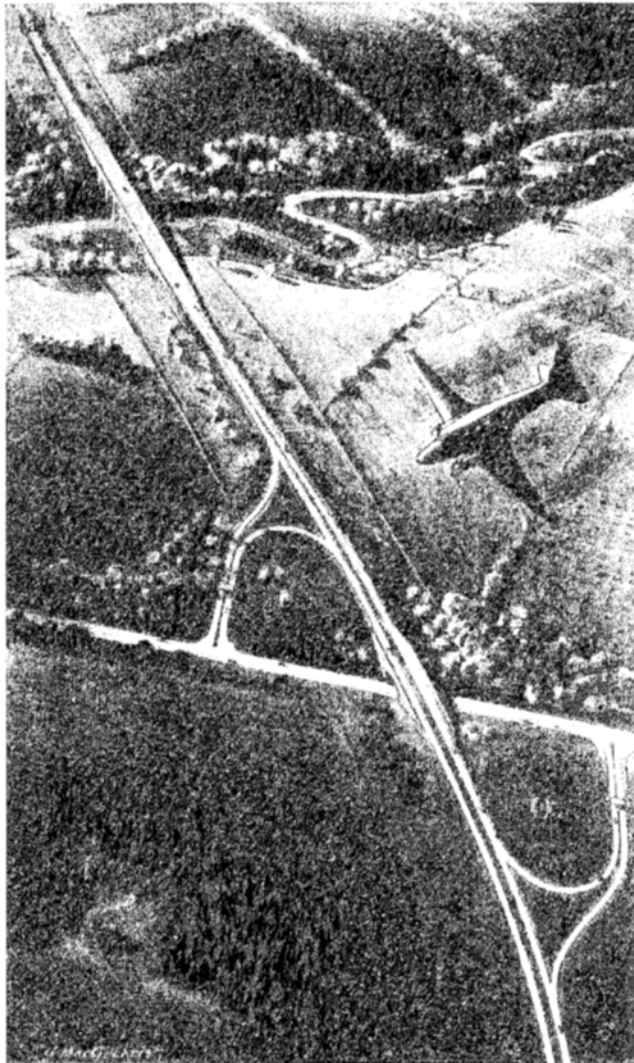


PLATE 32.—A typical grade separation, access roads, and toll booths for a two-lane road widened to three lanes at the approach to an access point by addition of a central lane of contrasting color and surface texture.



PLATE 31.—A typical grade separation, access roads, and toll booths for a four-lane road.



PLATE 20. —A four-lane toll highway in a suburban and urban area, showing relation to existing streets.

FIGURE 27. *Renderings of future American highway system, from Toll Roads and Free Roads (Public Roads Administration, 1939).*

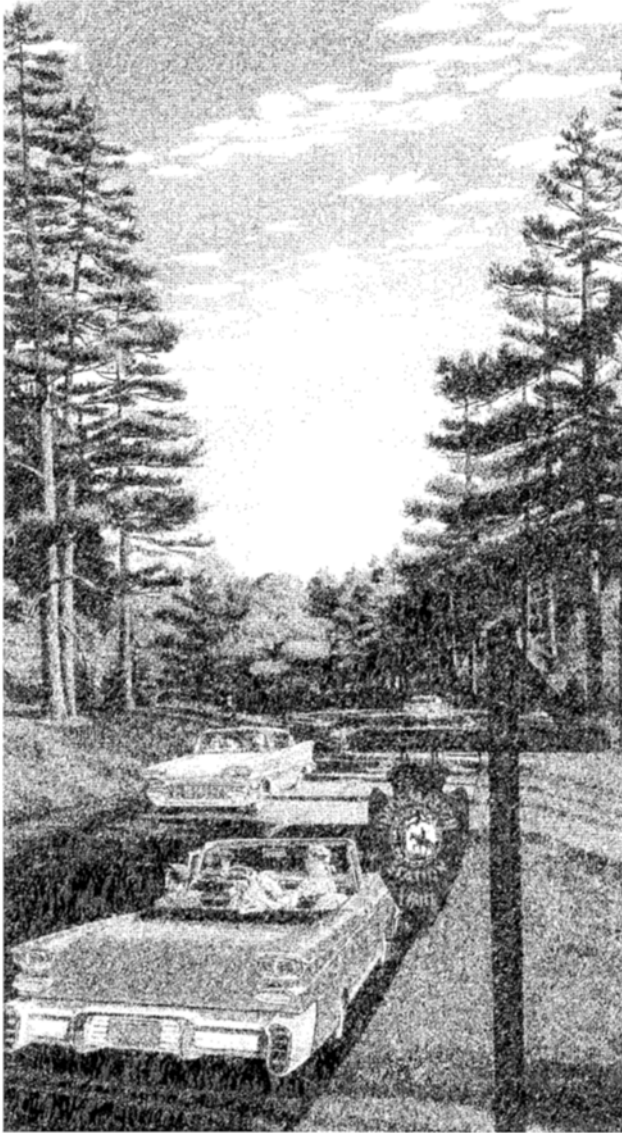


FIGURE 29. *Natchez Trace Parkway* (detail from 1959 oil company map: courtesy Sara Leach).



FIGURE 30. *Freeways* (vintage postcard: author).

less vulnerable to lawsuits by litigious parties intent on blaming outdated designs for accidents caused by inappropriate speeds and inattentive drivers.

Traditional parkways continued to have their place, of course, but as specialised landscapes to be enjoyed on carefully bracketed occasions, like a trip to the museum or a hike in the back country. With recreational practices tending toward more active physical exertion and environmental sensitivities growing by leaps and bounds in the 1950s and 1960s, motoring was no longer seen as the ideal way to experience America's scenic splendours. A number of proposed parkway developments were stopped in their tracks by environmentalist opposition (figure 32).³² Others, most prominently the Natchez Trace, languished for lack of funds, no longer able to attract the popular attention and political support they commanded during the parkway movement's heyday. While they were overshadowed by mainstream highway developments, the last generation of regionally oriented parkways demonstrated that it was possible to adapt classic parkway design techniques to higher speeds and traffic volumes. Completed during the 1950s and 1960s, the Taconic, Garden State, Palisades and Baltimore–Washington parkways might not have been as fast and efficient as conventional expressways and interstates,

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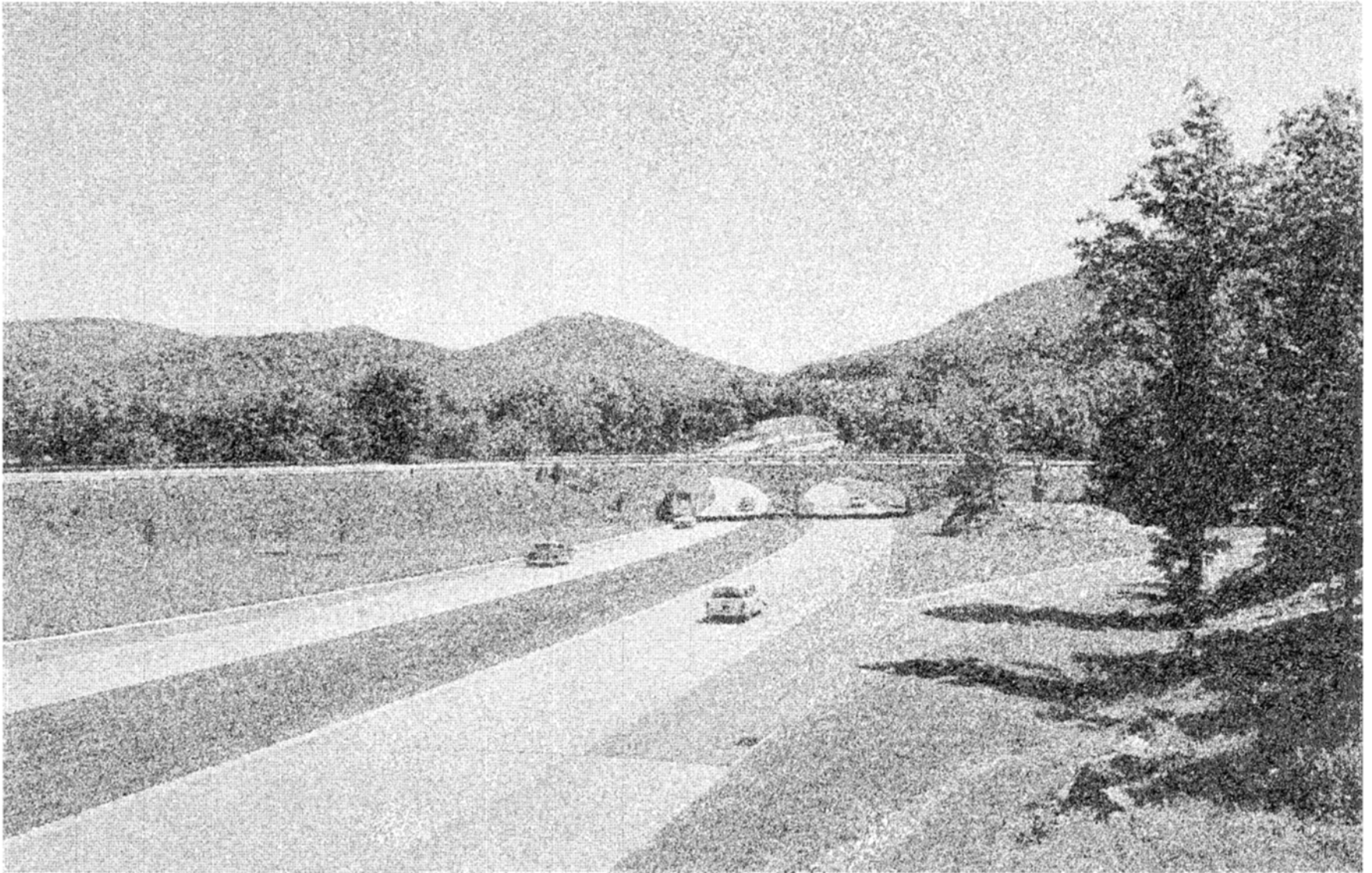


FIGURE 34. *Palisades Interstate Parkway, c. 1960 (vintage postcard: author).*

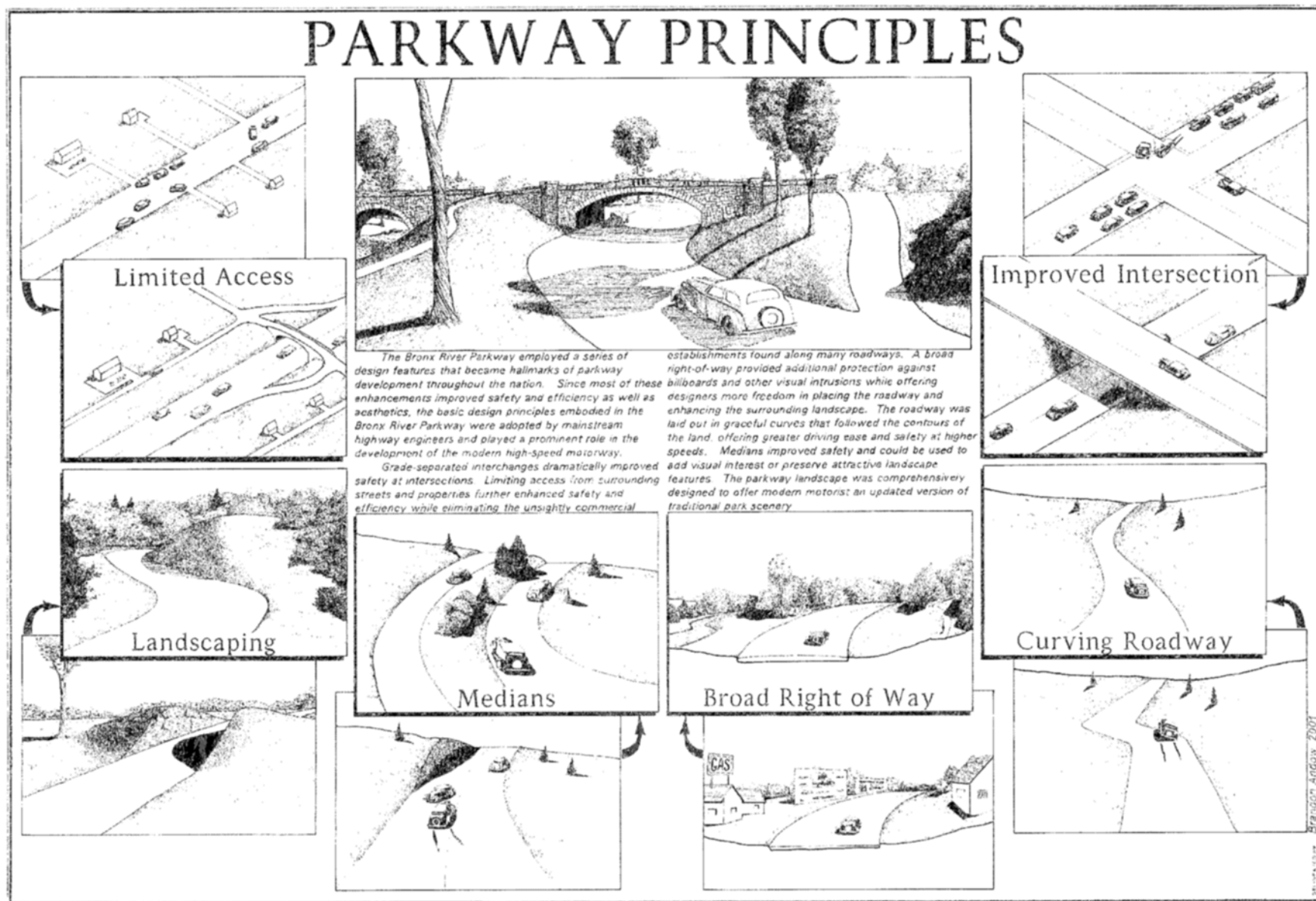


FIGURE 36. Expository drawing from documentation project on Bronx River Parkway (Historic American Engineering Record, 2001).

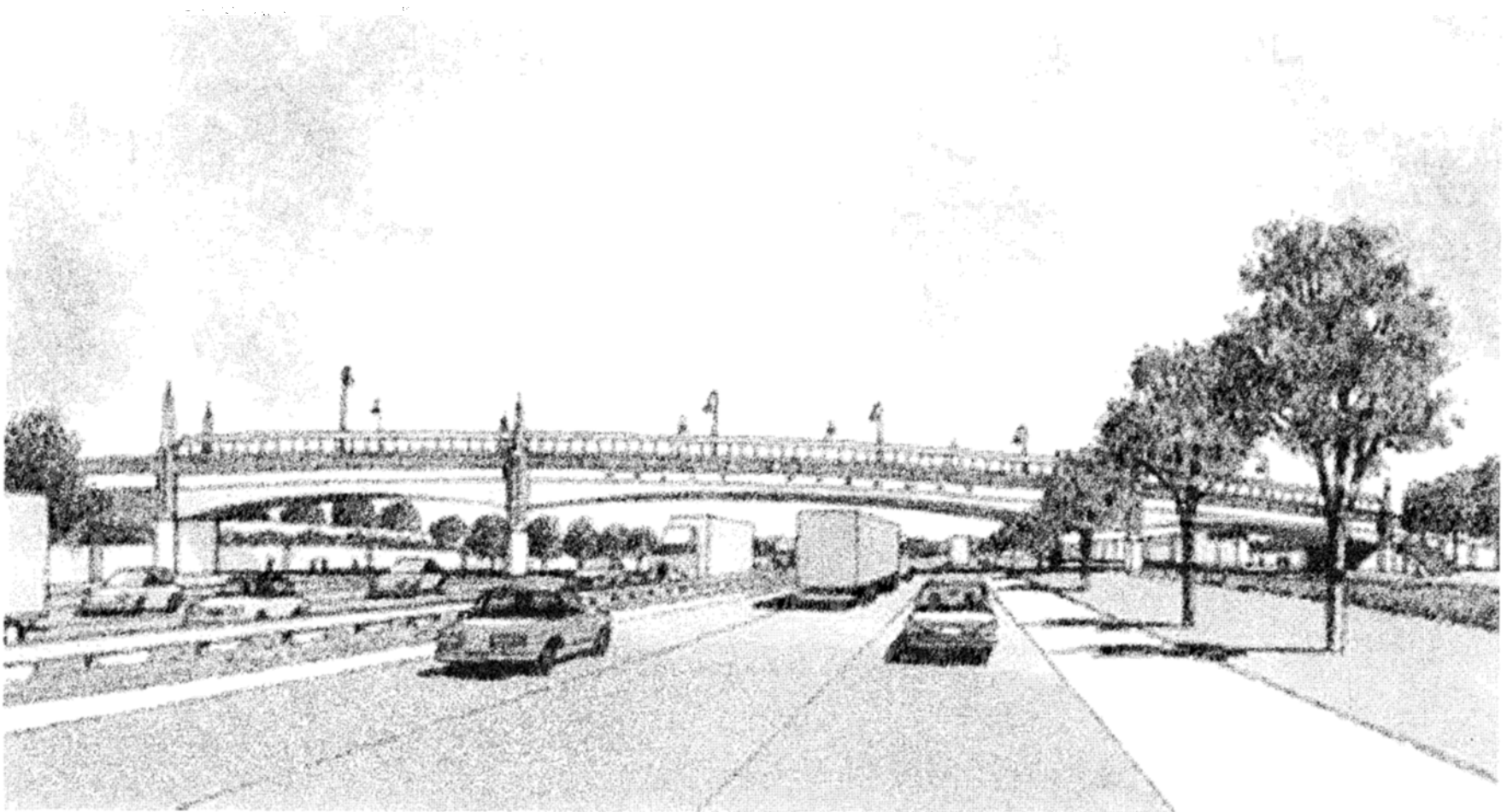


FIGURE 37. *Rendering of proposal for Kelly Parkway, San Antonio, Texas, 2004 (courtesy Parsons Brinckerhoff).*

NOTES

1. A comprehensive history of American parkway development has yet to be published. Brief discussions of the topic appear in several studies of American landscape architecture, city planning and highway engineering, which include: NORMAN T. NEWTON, *Design on the Land: The Development of Landscape Architecture* (Cambridge, 1971); CHRISTOPHER TUNNARD and BORIS PUSHKAREV, *Man-made America: Chaos or Control?* (New Haven, CT, 1963); and DAVID SCHUYLER, *The New Urban Landscape: The Redefinition of City Form in Nineteenth-century America* (Baltimore, MD, 1986). Numerous insights into parkway development are offered in CLAY MCSHANE, *Down the Asphalt Path: The Automobile and the American City* (New York, 1994). BRUCE RADDE, *The Merrit Parkway* (New Haven, CT, 1993) and SARA AMY LEACH, 'Fifty years of parkway construction in and around the nation's capital', in *Roadside America: The Automobile in Design and Culture*, ed. JAN JENNINGS (Ames, IA, 1990, pp. 185–197) also provide brief summaries of the genre. The author's dissertation provides an overview of American parkway history and examines the development of Mount Vernon Memorial Highway in detail (see TIMOTHY DAVIS, *Mount Vernon Memorial Highway and the Evolution of the American Parkway*. Doctoral dissertation, University of Texas–Austin, 1997). A summary of nineteenth-century parkway development and a chronicle of the history of a parkway that spanned both nineteenth- and twentieth-century eras can be found in TIMOTHY DAVIS, 'Rock Creek and Potomac Parkway, Washington, DC: The evolution of a contested urban landscape', *Studies in the History of Gardens and Designed Landscapes*, 19 (1999), pp. 123–237.
2. See WARREN SUSMAN, *Culture as History: The Transformation of American Society in the Twentieth Century* (New York, 1973); MICHAEL KAMMEN, *Mystic Chords of Memory: The Transformation of Tradition in American Culture* (New York, 1991); LAWRENCE LEVINE, *The Unpredictable Past: Essays in American Cultural History* (New York, 1993); JEFFREY MEIKLE, 'Domesticating modernity: Ambivalence and appropriation, 1920–1940', in *Designing Modernity: The Arts of Reform and Persuasion: Selections from the Wolfsonian*, ed. WENDY KAPLAN (London, 1995, pp. 143–167). The contemporary penchant for historic preservation, colonial revivalism and Anglo-centric public history has been examined in RICHARD GUY WILSON (ed.), *Re-creating the American Past: Essays on the Colonial Revival* (Charlottesville, VA, 2006); JAMES LINDGREN, *Preserving New England: Preservation, Progressivism and the Remaking of Memory* (New York, 1995); MICHAEL WALLACE, 'Visiting the past' and 'Reflections on the history of historic preservation', in *Presenting the Past: Essays on History and the Public*, ed. SUSAN PORTER BENSON, STEPHEN BRIER and ROY ROSENZWEIG (Philadelphia, PA, 1986), pp. 137–199; KARAL ANN MARLING, *George Washington Slept Here: Colonial Revivals and American Culture, 1876–1986* (Cambridge, MA, 1988); DAVID GEBHARD, 'The colonial revival in the 1930s', *Winterthur Portfolio*, 22 (1987), pp. 109–48; BRIDGET MAY, 'Progressivism and the colonial revival: The modern colonial house, 1900–1920', *Winterthur Portfolio*, 26 (1991), pp. 107–122.
3. See FREDERICK LAW OLNSTEIN and CALVERT VAUX, 'Report of the landscape architects and superintendents to the president of the board of commissioners of Prospect Park, Brooklyn', in *Eighth Annual Report of the Board of Commissioners of Prospect Park, Brooklyn, January 1868* (Brooklyn, NY, 1868). For the best overview of the relationship of late nineteenth-century parkway development to the broader evolution of American urban planning, see SCHUYLER, *New Urban Landscape*, pp. 126–146.
4. 'Unfit for modern motor traffic', *Fortune*, 14 (1936), pp. 85–99; *Life* magazine quoted in PHIL PATTON, *Open Road: A Celebration of the American Highway* (New York, 1986), pp. 66–67; *Saturday Evening Post* editorial quoted in 'Why not make travel as stimulating to the eye as it is to the speedometer?', *American City*, 41 (1929), p. 121; BENTON MACKAYE and LEWIS MUMFORD, 'Townless highways for the motorist: A proposal for the automobile age', *Harper's Monthly*, 163 (1931), pp. 347–356.
5. For a more detailed summary of American parkway development in the 1920s and 1930s, see DAVIS, *Mount Vernon Memorial Highway*, pp. 158–203. Contemporary accounts include: E. W. JAMES, 'Parkway features of interest to the highway engineer', *Public Roads*, 10 (1929), pp. 21–28; JAY DOWNER, 'How Westchester treats its roadsides', *American Civic Annual*, 1930, pp. 165–167; JAY DOWNER, 'Principles of Westchester's parkway system', *Civil Engineering*, 4 (1934), pp. 85–87; STANLEY ABBOT, 'Ten years of the Westchester County park system', *Parks and Recreation*, 16 (1933), pp. 305–314; GILMORE CLARKE, 'Modern motor arteries', in *Planning Problems of Town, City and Region: Papers and Discussion at the Twenty-second National Conference on City Planning, 23–26 June 1930* (Philadelphia, PA, 1930), pp. 61–75; GILMORE CLARKE, 'Is there a solution for the through traffic problem?', *Parks and Recreation*, 13 (1930), pp. 367–375; GILMORE CLARKE, 'Modern motorways', *Architectural Record* (December 1933), pp. 430–437.
6. Subsequent critics such as Robert Caro (*The Power Broker: Robert Moses and the Fall of New York* (New York, 1974)) have assailed parkway builders for this practice on the grounds that it reflected a reprehensible form of class discrimination. Such complaints may cater to current academic fads, but they reflect a poor command of history and geography.

- Park-makers long excluded such vehicles for the reasons outlined above. Contrary to Caro-induced myth, moreover, Long Island's Jones Beach has long been accessible by public transport: the Wantagh Parkway, which connected Jones Beach with the Long Island Railroad station at Wantagh, was equipped with higher overpasses that accommodated bus traffic, as can be seen in Figure 9. Contemporary accounts also note that many people hitchhiked along the parkways, though the practice was technically forbidden ('Robert (Or-I'll-Resign) Moses', *Fortune* (June 1938), p. 124).
7. This vein of criticism can be seen in a series of published responses to a Gilmore Clarke presentation at the Twenty-second National Conference on City Planning in June 1930 (National Conference on City Planning, *Planning Problems of Town, City and Region* (Philadelphia, PA, 1930), pp. 67-75).
 8. United States Department of Agriculture (USDA), *Roadside Improvement: US Department of Agriculture Miscellaneous Publication 191* (Washington, DC, 1934); republished in Germany as: WILBUR SIMONSON and R. E. ROYALL, *Landschaftsgestaltung an der Strasse* (Berlin, 1935).
 9. USDA, Bureau of Public Roads, *The Mount Vernon Memorial Highway: History, Design and Progress in Construction* (Washington, DC, 1930). Italian coverage of the Memorial Highway includes: I. VANDONE, 'La Strada Commerativa da Washington a Mount Vernon', *Strade*, 13 (1931), pp. 2-8; Consiglio Nazionale Delle Ricerche Comitato per L'Ingegneria, 'La Strada Commerativa da Washington a Mount Vernon', in *La Partecipazione Italiana al Sesto Congresso Internazionale della Strade, Washington, October 1930* (Rome, 1931), pp. 229-233.
 10. STANLEY ABBOTT, 'Parkways — Past, present and future', *Parks and Recreation*, 31 (1948), p. 683; GILMORE CLARKE, 'Beauty: A wanting factor in the turnpike design', *Landscape Architecture*, 32 (1942), pp. 53-54. Clarke underscored the American parkway's seminal status in: GILMORE CLARKE, 'Westchester parkways: An American development in landscape architecture', *Landscape Architecture*, 28 (1937), pp. 318-321.
 11. 'Building the Mount Vernon Memorial Highway', 31 min., si., b & w, 35 mm (1933); USDA Films, Record Group 33, Moving Pictures Branch, National Archives.
 12. United States Congress, House Committee on Roads, *Roads: Hearings before the Committee on Roads ... on H.R. 524* (68th Congress, 1st Session, 25 April 1924), pp. 16-21.
 13. *American Highways*, 12 (1933).
 14. H. S. FAIRBANK, 'Modern design characterizes highway memorial to Mount Vernon', *American City*, 43 (1930), pp. 149-150; WILBUR SIMONSON, 'The Mount Vernon Memorial Highway: Most modern motorway, designed as memorial to country's first president, now under construction', *American City*, 43 (1930), pp. 85-88; 'A notable highway', *Engineering News-Record*, 23 July 1931, p. 122; 'The Mount Vernon Memorial Highway', *Engineering News-Record* 23 July 1931, pp. 124-127; 'The Mount Vernon Memorial Highway', *American Highways*, 11 (1932), pp. 11-13; R. E. TOMS and J. W. JOHNSON, 'The design and construction of the Mount Vernon Memorial Highway', *Journal of the American Concrete Institute*, 4 (1932), pp. 563-584; R. E. ROYALL, 'The Mount Vernon Memorial Highway', *The Military Engineer*, 24 (1932), pp. 238-242.
 15. GILMORE CLARKE, 'Mount Vernon Memorial Highway', *Landscape Architecture*, 22 (1932), pp. 179-189; R. E. TOMS, 'Design and construction of Mount Vernon Memorial Highway', *Parks and Recreation*, 15 (1932), pp. 537-544.
 16. 'Blow the horns for Mount Vernon's superhighway', *Literary Digest*, 9 January 1932, pp. 42-43.
 17. 'The Mt. Vernon Memorial Highway', *American Motorist*, 6 (1932), pp. 21, 28.
 18. ABBOTT, 'Parkways — Past, present and future', p. 684.
 19. For more on the contrived nature of parkway landscapes and the tensions between the goals of institutionally empowered designers and local or vernacular interests, see BARRY M. BUXTON, *Mabry Mill Historic Resource Study* (Washington, DC, 1987); PHIL NOBLITT, 'The Blue Ridge Parkway and the myths of the pioneer', *Appalachian Journal*, 21 (1994), pp. 394-409; TIMOTHY DAVIS, "'A pleasant illusion of unspoiled countryside': The American parkway and the problematics of institutional vernacular', in *Constructing Image, Identity and Place: Perspectives in Vernacular Architecture*, IX, eds ALISON K. HOAGLAND and KENNETH A. BREISCH (Knoxville, TN, 2003), pp. 228-246; KATHLEEN LAFRANK, 'Real and ideal landscapes along the Taconic State Parkway', in HOAGLAND and BREISCH, *Constructing Image, Identity and Place*, pp. 247-262.
 20. For the classic articulation of the concept of 'invented traditions', see ERIC HOBSBAWM and TERENCE RANGER (eds), *The Invention of Tradition* (New York, 1983). The concept of 'romantic nationalism' and its material expressions are discussed in: WENDY KAPLAN, 'Traditions transformed: Romantic nationalism in design, 1890-1920', in KAPLAN, *Designing Modernity*, pp. 19-47. The tension between modernity and anti-modernity is explicitly addressed in: MEIKLE, 'Domesticating Modernity'. Various manifestations of these phenomena in early twentieth-century America are explored in: JOHN BODNAR, *Remaking America: Public Memory, Commemoration and Patriotism in the Twentieth Century* (Princeton, NJ, 1992).
 21. CLEVELAND RODGERS, 'Highways and parkways', *The Studio*, 127 (1944), pp. 204-208.
 22. 'Robert (Or-I'll-Resign) Moses', pp. 70-78, 124-141.
 23. 'Along the Bronx River,' postcard published by the Ruben Publishing Co., Newburgh, NY, c. 1935 (author's collection).
 24. SIGFRIED GIEDION, *Space, Time and Architecture: The Growth of a New Tradition* (Cambridge, 1941), pp. 550-559.
 25. For further analysis of the cultural resonance of streamlining in America in the 1920s and 1930s, see JEFFREY MEIKLE, *Twentieth Century Limited: Industrial Design in America, 1925-1939* (Philadelphia, PA,

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- 1979), esp. pp. 135–187. Terry Smith recapitulates Meikle's observations and adds some interpretations of his own in: TERRY SMITH, *Making the Modern: Industry, Art and Design in America* (Chicago, IL, 1993), pp. 353–384.
26. The 'friction'-based traffic theories of Dr Miller McClintock, head of Harvard's Bureau of Street Traffic Research, were summarised in 'Unfit for modern motor traffic', pp. 85–99.
27. WILBUR SIMONSON, 'The Mount Vernon Memorial Highway Unit VI: Final Report: The Landscape Architectural Problems in its Development,' United States Bureau of Public Roads, United States Department of Agriculture, 1932 (illustrated copy in United States Department of Transportation Library, Washington, D.C.), pp. 17–18, 197, captions to plates 70, 79.
28. EDWARD BASSETT, 'The freeway: A new kind of thoroughfare', *American City*, 42 (1930), p. 95.
29. A comprehensive, profusely illustrated overview of the Futurama, along with an elucidation of its underlying principles, is provided in: NORMAN BEL GEDDES, *Magic Motorways* (New York, 1940). Futurama's significance is discussed in many sources including: MEIKLE, *Twentieth Century Limited*; SMITH, *Making the Modern*.
30. United States Public Roads Administration. *Toll Roads and Free Roads* (Washington, DC, 1939), pp. 41–52, plates 30–33, 54–55.
31. 'Streamlined mountain thoroughfare', *Better Roads*, 10 (1940), pp. 13–14, 31–32; RALPH STEWART, 'The Pennsylvania Turnpike and its landscape treatment', *Landscape Architecture*, 32 (1942), pp. 47–52; CLARKE, 'Beauty', pp. 53–54; MALCOLM DILL, 'The cross-section: Its effect on safety and economy', *Landscape Architecture*, 32 (1942), pp. 52–53; RALPH GRISWOLD, 'Precedent for superhighway landscape yet to be found', *Landscape Architecture*, 32 (1942), pp. 54–55; LAURIE COX, 'Appearance: Essential element in highway design', *Landscape Architecture*, 32 (1942), pp. 55–56.
32. One successful anti-parkway protest is chronicled by former National Park Service historian Barry Mackintosh in 'Shoot out on the old C & O Canal: The great parkway controversy, 1950–1960', *Maryland Historical Magazine*, 90 (1995), pp. 140–163).
33. For an overview of this project consisting of 331 measured and interpretive drawings and brief summary of national park road history, see TIMOTHY DAVIS, TODD CROTEAU and CHRISTOPHER MARTSON (eds), *America's National Park Roads and Parkways: Drawings from the Historic American Engineering Record* (Baltimore, MD, 2004).